

DSA Test 3 – Set A

Duration: 60 minutes
Total Questions: 5

Question 1 – Pattern (Easy)

Print a right-angled triangle of stars.

Input: `n = 5`

Output:

```
*
* *
* * *
* * * *
* * * * *
```

Question 2 – Pattern (Easy)

Print a reverse number triangle.

Input: `n = 5`

Output:

```
5 4 3 2 1
5 4 3 2
5 4 3
5 4
5
```

Question 3 – Hashing (Easy)

Frequency of Each Element

Given an array of `n` integers, print the frequency of each distinct element.

Input:

```
n = 7
arr = [1, 2, 2, 3, 3, 3, 4]
```

Output:

```
1 → 1
2 → 2
3 → 3
4 → 1
```

Constraints: Use a HashMap. $O(n)$ time.

Question 4 – Sorting Algorithm (Easy)

Implement Bubble Sort

Given an array of `n` integers, sort it in ascending order using Bubble Sort. Do NOT use built-in sort functions.

Input:

```
n = 6
arr = [64, 34, 25, 12, 22, 11]
```

Output: [11, 12, 22, 25, 34, 64]

Expected complexity: $O(n^2)$ time, $O(1)$ space.

Question 5 — Arrays (Easy)

Left Rotate an Array by 1 Position

Given an array of `n` integers, rotate it to the left by one position.

Input:

```
n = 5
arr = [1, 2, 3, 4, 5]
```

Output: [2, 3, 4, 5, 1]

Constraints: $O(n)$ time, $O(1)$ space.

Good luck!